

<!DOCTYPE html>

AGENTHINKMESH — SYSTEM ARCHITECTURE

Data flow diagram · Actual implementation only · No marketing language

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<!-- Layer 1: Client Browser -->
<div class="layer layer-client">
  <div class="layer-label">Layer 1</div>
  <div class="layer-name">Client Browser</div>
  <div class="layer-detail">React 19 · Tailwind CSS 4 · tRPC client
hooks</div>
  <div class="layer-detail">User submits task prompt via UI
form</div>
</div>

<div class="arrow">
  <div class="arrow-line"></div>
  <div class="arrow-head"></div>
  <div class="arrow-label">HTTP POST · JSON payload</div>
</div>

<!-- Layer 2: tRPC API Layer -->
<div class="layer layer-trpc">
  <div class="layer-label">Layer 2</div>
  <div class="layer-name">tRPC API Layer</div>
  <div class="layer-detail">server/routers.ts · Express 4 ·
/api/trpc</div>
  <div class="layer-detail">Auth context injected · Input validated
via Zod</div>
</div>

<div class="arrow">
  <div class="arrow-line"></div>
  <div class="arrow-head"></div>
  <div class="arrow-label">Validated procedure call</div>
</div>

<!-- Layer 3: councilEngine.ts -->
<div class="layer layer-engine">
  <div class="layer-label">Layer 3</div>
  <div class="layer-name">councilEngine.ts</div>
  <div class="layer-detail">Orchestrates parallel evaluation
calls</div>
  <div class="badge">10 × Promise.all() – concurrent LLM calls</div>
</div>

<div class="arrow">
  <div class="arrow-line"></div>
  <div class="arrow-head"></div>
  <div class="arrow-label">10 concurrent HTTPS requests</div>

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</div>

<!-- Layer 4: LLM API -->
<div class="layer layer-llm">
  <div class="layer-label">Layer 4</div>
  <div class="layer-name">LLM API Layer</div>
  <div class="layer-detail">Anthropic Claude Sonnet (claude-3-5-sonnet-20241022)</div>
  <div class="layer-detail">server/_core/llm.ts · invokeLLM() helper</div>
  <div class="badge badge-green">External API · Bearer auth · No key exposed to client</div>
</div>

<div class="arrow">
  <div class="arrow-line"></div>
  <div class="arrow-head"></div>
  <div class="arrow-label">10 × structured JSON responses</div>
</div>

<!-- Layer 5: Responses returned -->
<div class="layer layer-response">
  <div class="layer-label">Layer 5</div>
  <div class="layer-name">Responses returned to councilEngine</div>
  <div class="layer-detail">All 10 agent outputs collected via Promise.all()</div>
  <div class="layer-detail">Each response: { agentLabel, result, confidence }</div>
  <div class="badge badge-green">Awaited in parallel · Timeout handled per call</div>
</div>

<div class="arrow">
  <div class="arrow-line"></div>
  <div class="arrow-head"></div>
  <div class="arrow-label">Aggregated output array</div>
</div>

<!-- Layer 6: Consensus Logic -->
<div class="layer layer-consensus">
  <div class="layer-label">Layer 6</div>
  <div class="layer-name">Consensus Logic</div>
  <div class="layer-detail">Server-side only · Never exposed to client</div>
  <div class="layer-detail">Conflict detection · Confidence scoring · Summary generation</div>

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    <div class="badge badge-orange">mesh.summariseOutputs procedure .
    invokeLLM() synthesis call</div>
</div>

<div class="arrow">
    <div class="arrow-line"></div>
    <div class="arrow-head"></div>
    <div class="arrow-label">Structured deal record</div>
</div>

<!-- Layer 7: MySQL Database -->
<div class="layer layer-db">
    <div class="layer-label">Layer 7</div>
    <div class="layer-name">MySQL Database – deal storage</div>
    <div class="layer-detail">TiDB · Drizzle ORM ·
drizzle/schema.ts</div>
    <div class="layer-detail">Tables: taskHistory · agents ·
agentMetrics · documents · users</div>
    <div class="badge badge-cyan">Stores: task, agentsUsed,
executionTime, outputs, userId</div>
</div>

<div class="arrow">
    <div class="arrow-line"></div>
    <div class="arrow-head"></div>
    <div class="arrow-label">Persisted record retrieved</div>
</div>

<!-- Layer 8: Report Generation + PDF -->
<div class="layer layer-report">
    <div class="layer-label">Layer 8</div>
    <div class="layer-name">Report Generation + PDF Output</div>
    <div class="layer-detail">Client-side: window.print() → browser PDF
renderer</div>
    <div class="layer-detail">Includes: all agent outputs + Final
Summary card</div>
    <div class="badge badge-pink">HTML → PDF via browser print API · No
server-side PDF library</div>
</div>

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